NATIONAL ECONOMICS UNIVERSITY (NEU)

Faculty of Mathematical Economics

RESEARCH REPORT — TIME SERIES ANALYSIS

Daily Temperature Forecasting Using Time Series and Machine Learning Models

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Abstract

This paper analyzes and forecasts daily temperature in **Mumbai, India** using meteorological data recorded over the period 2016–2020 (1,781 observations). Three models are estimated and compared: a univariate SARIMA model, a multivariate SARIMAX with exogenous variables, and an XGBOOST gradient-boosting model. Exogenous variables for SARIMAX are selected through Granger causality testing, identifying wind direction, sea-level pressure, humidity, and dew point as statistically significant predictors. On the held-out test set (January–November 2020), SARIMAX achieves the highest forecast accuracy ($R^2 = 0.976$, RMSE $= 0.313$ °C, MAPE $= 0.83\%$, 95% CI coverage $= 94.7\%$), outperforming both XGBOOST ($R^2 = 0.938$) and SARIMA ($R^2 = 0.836$). Results underscore the importance of causality-based variable selection for multivariate time series forecasting.

Keywords: time series forecasting, SARIMA, SARIMAX, XGBoost, Granger causality, Mumbai temperature prediction, confidence interval.

Contents

1. Introduction

Accurate temperature forecasting is essential for a wide range of economic and social applications, including agricultural planning, energy demand management, disaster preparedness, and public health. Mumbai, India — a tropical coastal city characterized by a distinct monsoon season (June–September) and strong annual temperature cycles — provides a particularly suitable case study for seasonal time series forecasting. Despite rapid advances in numerical weather prediction, statistical and machine learning approaches remain valuable for short-range forecasting at the local level, particularly when historical observation data are available but computational resources for physical models are limited.

This paper addresses the question: *which combination of model specification and variable selection best forecasts daily temperature using historical meteorological records?* The analysis makes three contributions. First, it applies Granger causality testing to identify exogenous predictors in a principled, data-driven manner, avoiding the common practice of selecting variables based on pairwise correlation alone. Second, it compares two statistical time series models (SARIMA and SARIMAX) with a widely used machine learning approach (XGBOOST), providing a comprehensive benchmark. Third, it uses a rolling one-step-ahead forecast methodology for SARIMA and SARIMAX to produce realistic out-of-sample forecasts, accompanied by 95% prediction intervals to quantify forecast uncertainty.

The empirical results show that SARIMAX with Granger-selected variables substantially outperforms the univariate SARIMA baseline (RMSE reduction of 61.5%) and also outperforms XGBOOST, despite relying on fewer inputs and no non-linear interaction terms. The 95% confidence interval achieves a coverage rate of 94.7%, close to the nominal level.

The remainder of the paper is organized as follows. Section 2 reviews related literature. Section 3 describes the data and methodology. Section 4 reports empirical results. Section 5 discusses findings and limitations, and Section 6 concludes.

2. Literature Review

2.1. Statistical Time Series Models

Time series forecasting of meteorological variables has a long tradition in statistics and atmospheric science. The Box–Jenkins framework (Box et al., 2015) provides a systematic approach to modeling univariate time series through ARIMA models. The seasonal extension, SARIMA, captures periodic patterns through multiplicative seasonal differencing and seasonal AR/MA components, making it well suited to variables such as temperature that exhibit strong annual cycles (Chatfield, 2000).

Multivariate extensions, including SARIMAX, incorporate exogenous variables to improve forecast accuracy when external predictors carry information beyond the target series' own history (Hamilton, 1994). The performance of these models is sensitive to variable selection: including irrelevant predictors introduces multicollinearity and may worsen accuracy relative to the univariate baseline.

2.2. Granger Causality and Variable Selection

Granger (1969) introduced the concept of Granger causality as a formal test of predictive precedence: variable X is said to Granger-cause Y if past values of X improve the forecast of Y beyond what past values of Y alone can achieve. Formally, X Granger-causes Y if:

$$\sigma^2\left[\hat{Y}_{t+1} \mid \mathcal{F}_t(X, Y)\right] < \sigma^2\left[\hat{Y}_{t+1} \mid \mathcal{F}_t(Y)\right] \tag{1}$$

This criterion provides a principled basis for exogenous variable selection in SARIMAX, distinguishing genuine predictive relationships from spurious correlations driven by shared seasonal patterns. Granger causality tests have been widely used in macroeconometrics (Hamilton, 1994) but their application to meteorological variable selection for SARIMAX forecasting remains limited in the literature.

2.3. Machine Learning for Time Series Forecasting

Gradient-boosting algorithms, particularly XGBOOST (Chen and Guestrin, 2016), have demonstrated strong predictive performance across tabular forecasting tasks. For time series applications, these models are typically augmented with lag features and rolling statistics. XGBOOST's ability to capture non-linear interactions makes it a natural benchmark for comparison with statistical models. Comparative studies generally find that machine learning models improve relative to statistical approaches as the number of observations and features grows, while statistical models retain advantages in low-data settings and for interpretability (Chatfield, 2000).

3. Data and Methodology

3.1. Data Description

The dataset consists of 1,781 daily meteorological observations recorded in Mumbai, India between 1 January 2016 and 15 November 2020. The target variable is `temp`, the daily mean temperature in degrees Celsius. Seven candidate predictor variables are available, as described in Table 1. Descriptive statistics are reported in Table 2. Temperature exhibits clear seasonal variation, with monthly means ranging from 25.4 °C in January to 30.8 °C in May (range: 5.4 °C). The distribution is slightly left-skewed ($\gamma_1 = -0.51$) with near-normal kurtosis ($\gamma_2 = 0.16$). The linear trend is negligible ($+0.067$ °C/year, $R^2 = 0.002$).

Table 1: Data Dictionary

Variable	Type	Unit	Description
`temp`	float	°C	Daily mean temperature (target)
`dew`	float	°C	Dew point temperature
`humidity`	float	%	Relative humidity
`sealevelpressure`	float	hPa	Sea-level atmospheric pressure
`winddir`	float	°	Wind direction
`solarradiation`	float	—	Solar radiation
`windspeed`	float	km/h	Wind speed
`preciptype`	binary	0/1	Precipitation occurrence

Table 2: Descriptive Statistics (2016–2020, $n = 1{,}781$)

Variable	Mean	Std	Min	Median	Max
temp (°C)	28.34	1.96	20.20	28.50	32.80
dew (°C)	21.67	3.79	5.00	22.40	27.50
humidity (%)	69.58	15.22	28.60	70.30	98.20
sealevelpressure (hPa)	1008.91	3.12	994.10	1009.10	1017.40
winddir (°)	202.44	57.85	65.60	207.10	316.00
solarradiation	230.14	62.41	52.80	232.50	330.90
windspeed (km/h)	22.13	6.89	9.40	21.30	128.10
preciptype (0/1)	0.456	0.498	0	0	1

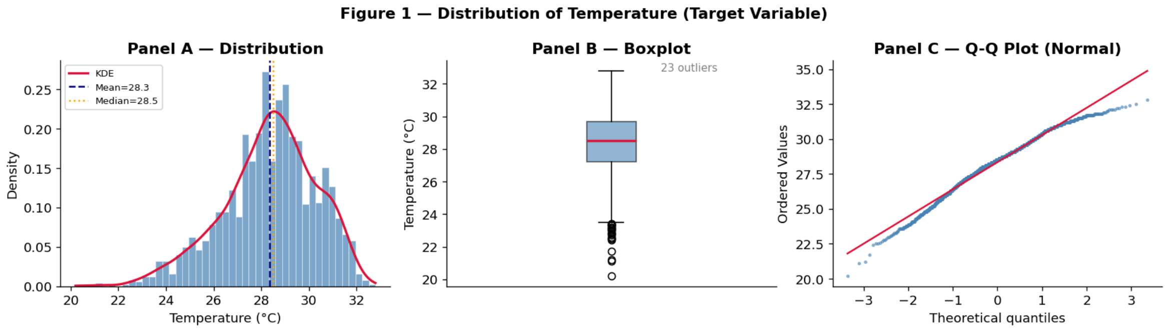


Figure 1: Distribution of daily temperature. Panel A: histogram with KDE overlay. Panel B: boxplot (23 outliers). Panel C: Q–Q plot.

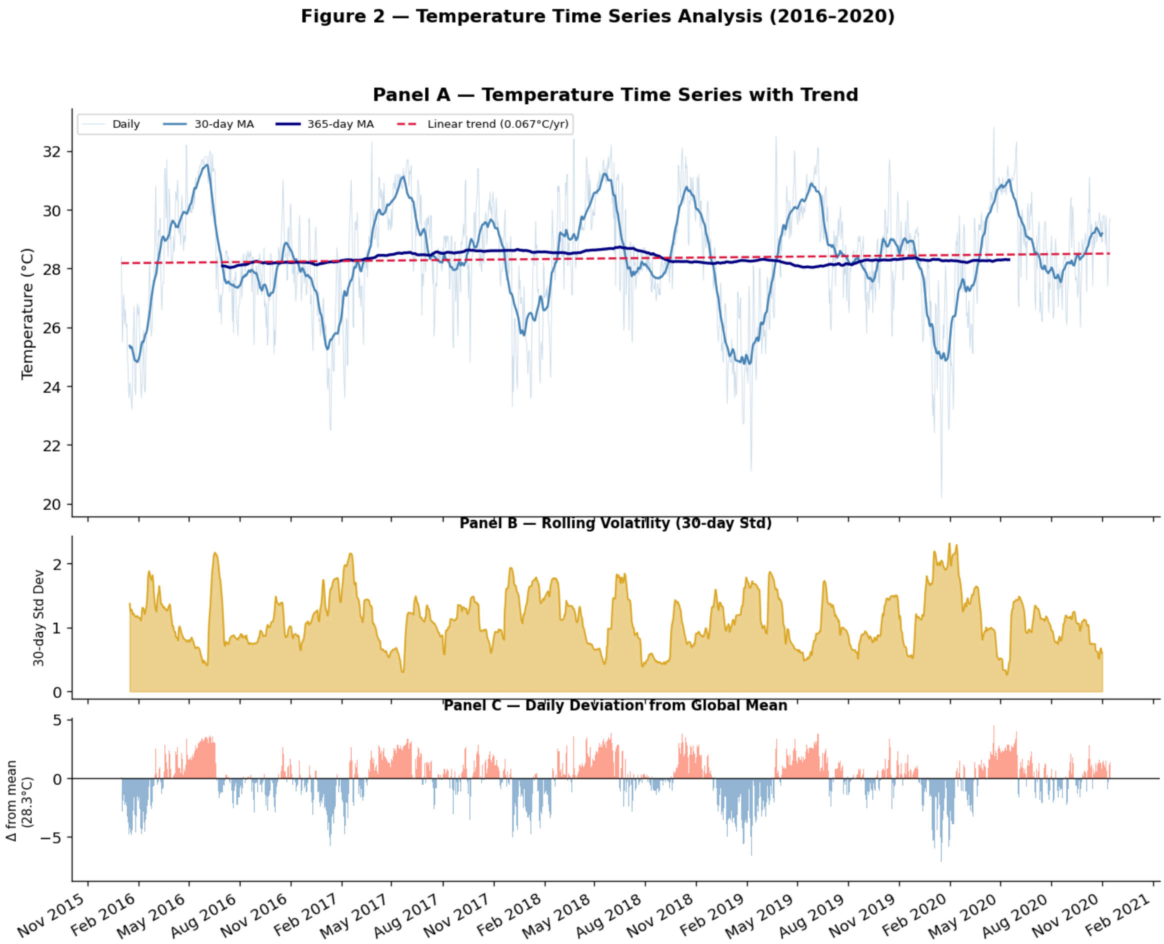
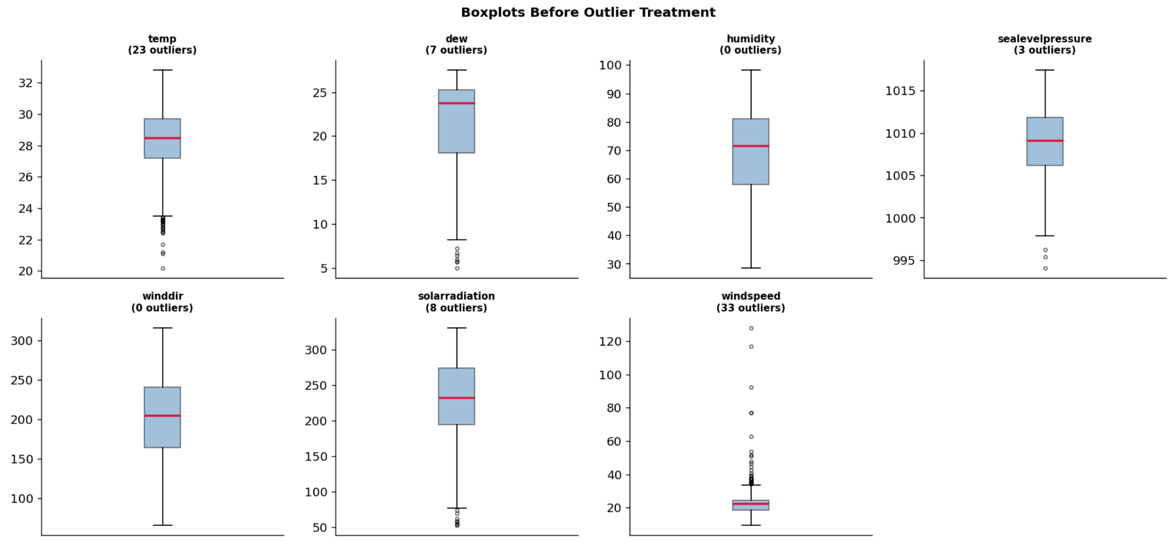


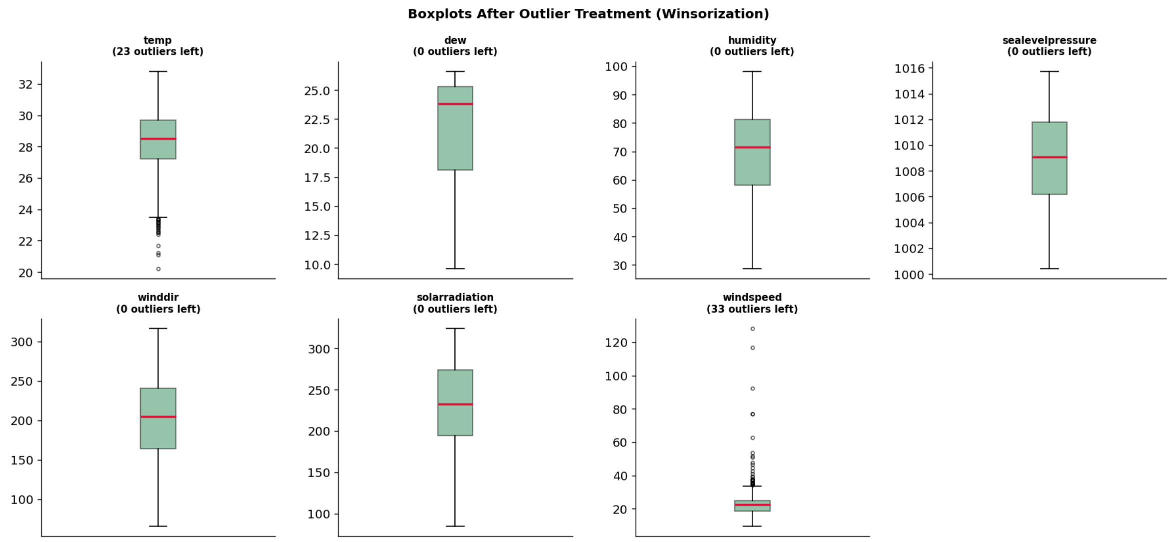
Figure 2: Temperature time series (2016–2020). Panel A: daily values with moving averages and linear trend. Panel B: rolling volatility. Panel C: deviation from global mean.

3.2. Preprocessing

The dataset contains no duplicate rows, no duplicate timestamps, and no missing values. Outlier detection via the IQR method is applied to all variables. Winsorization at the 1st–99th percentiles is applied to `solarradiation`, `sealevelpressure`, and `dew`. Temperature outliers (23 cold-season readings in January–February) and wind speed outliers (33 monsoon-season values, June–September) are retained as genuine meteorological events.



(a) Before Winsorization



(b) After Winsorization

Figure 3: Boxplots before and after outlier treatment.

3.3. Feature Engineering

Features are constructed in four groups targeting specific model classes (Table 3). To prevent data leakage: (1) lag features use `shift(1)`, ensuring only past values are used; (2) rolling statistics use `shift(1).rolling(w)`, excluding the current observation from the window; (3) StandardScaler parameters are fitted exclusively on the training set and applied to both sets. Exogenous variables for SARIMAX and features for XGBOOST are scaled independently to prevent cross-contamination.

Table 3: Feature Engineering Summary by Model

Group	Features	SARIMAX	XGBoost
Time features	month, quarter, dayofyear, season, sin / cos(doy), sin / cos(month)	×	✓
Exogenous lags	dew_lag1, sealevelpressure_lag1	✓	✓
Target lags	temp_lag1 ($r = 0.908$), temp_lag7 ($r = 0.630$)	×	✓
Rolling mean	temp_rollmean7 ($r = 0.801$)	×	✓
Total		**2**	**20**

SARIMA uses no external features.

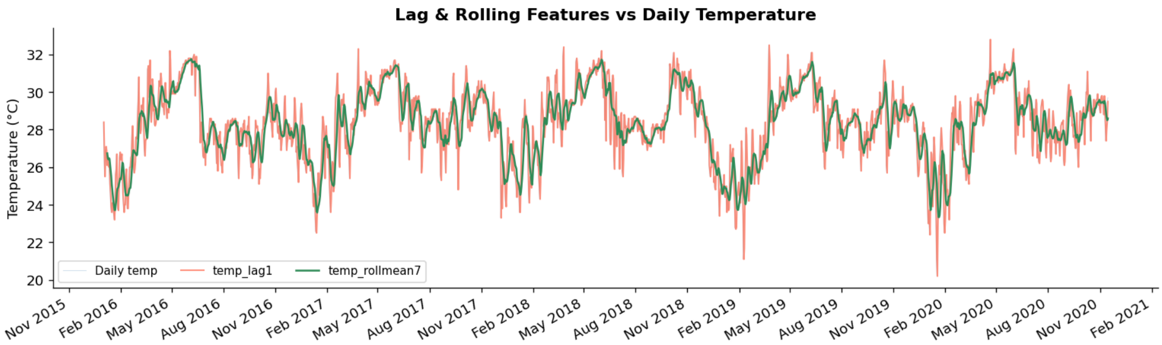


Figure 4: Lag and rolling features vs daily temperature. `temp_lag1` (red) tracks the series with a one-day offset; `temp_rollmean7` (green) provides a smoothed local trend.

3.4. Stationarity Testing

The Augmented Dickey–Fuller (ADF) test (Dickey and Fuller, 1979) is applied to assess the integration order of the temperature series (Table 4). The original series is stationary in levels at the 1% significance level (ADF $= -4.617$, $p < 0.001$). After first differencing the statistic falls to -16.391 ($p < 0.001$), confirming $d = 1$ is more than sufficient.

Table 4: ADF Test Results

Series	ADF stat.	p-value	CV 1%	CV 5%	Decision
Original (y_t)	-4.617	0.0001	-3.434	-2.863	Stationary
First diff. (Δy_t)	-16.391	< 0.001	-3.434	-2.863	Stationary

The ACF and PACF of the first-differenced series (Figure 5) show significant spikes at lags 1–3 in both functions (ACF lags 1–5: $[0.056, -0.130, -0.193, -0.087, -0.028]$; 95% CI: ± 0.047), suggesting candidate orders $p, q \in \{1, 2, 3\}$.

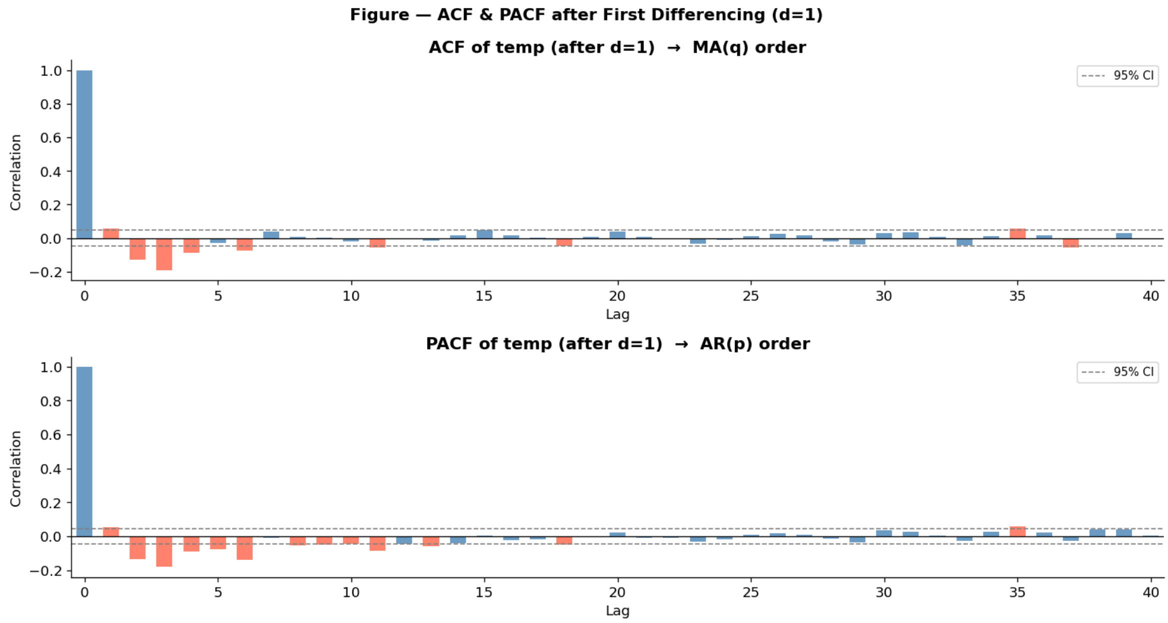


Figure 5: ACF and PACF of temperature after first differencing ($d = 1$). Red bars exceed the 95% CI (± 0.047). Significant spikes at lags 1–3 guide the SARIMA grid search.

3.5. Granger Causality Testing

All seven candidate exogenous variables are subjected to pairwise Granger causality tests with respect to temperature, using up to 7 lags on first-differenced series. A VAR model is estimated first; AIC selects lag order 4, validating 7 as a conservative upper bound. Results are reported in Table 5 and Figure 6.

Table 5: Granger Causality Test Results (H_0: Variable does not Granger-cause `temp`)

Variable	Best lag	Min p-value	Sig.	Decision
winddir	1	< 0.001	***	Granger-causes temp
humidity	3	0.0063	**	Granger-causes temp
sealevelpressure	1	0.0071	**	Granger-causes temp
dew	1	0.0315	*	Granger-causes temp
solarradiation	5	0.5802	ns	No effect
windspeed	3	0.2256	ns	No effect
preciptype	6	0.0720	ns	No effect

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns = not significant

Notably, `winddir` is the most significant Granger-cause ($p < 0.001$) despite a modest Pearson correlation with temperature ($r = 0.23$), illustrating the value of causality-based over correlation-based variable selection. Conversely, `solarradiation` ($r = 0.22$) fails the Granger test ($p = 0.58$), suggesting its correlation is driven by shared seasonal patterns already absorbed by the ARIMA structure.

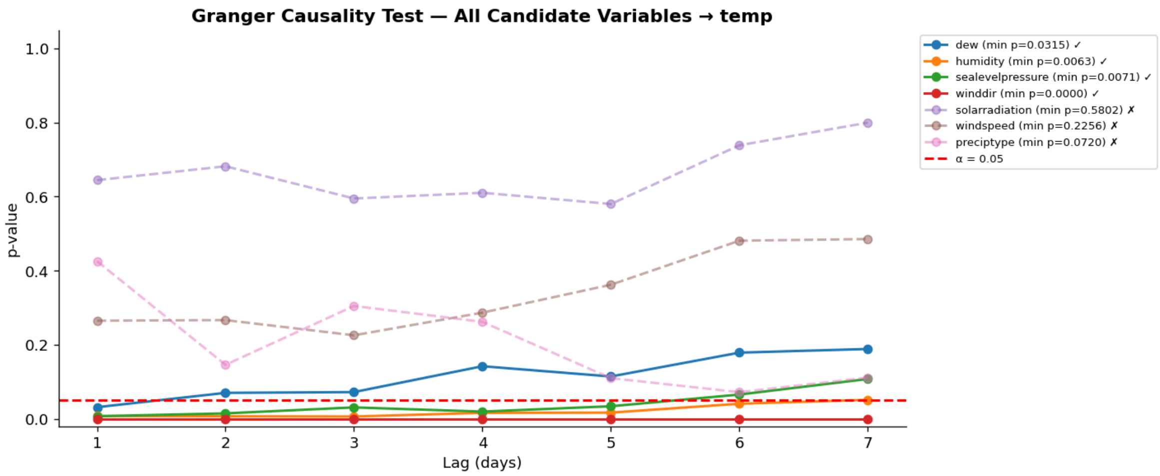


Figure 6: Granger causality p-values by lag. Solid lines: significant at $\alpha = 0.05$. Dashed lines: non-significant. Red line: $\alpha = 0.05$ threshold.

3.6. Model Specification

3.6.1. SARIMA

The $\text{SARIMA}(p, d, q)(P, D, Q)_s$ model is:

$$\Phi_P(B^s)\,\phi_p(B)\,\Delta^d\Delta_s^D\,y_t = \Theta_Q(B^s)\,\theta_q(B)\,\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2) \tag{2}$$

with $s = 12$, $d = D = 1$. Although the data are at daily frequency, $s = 12$ is adopted as a computational approximation of the annual seasonal cycle. Using $s = 365$ on daily data is computationally prohibitive for rolling forecast (320 re-fits per model). The annual seasonal pattern is additionally captured via Fourier $\sin$ / $\cos$ features in XGBoost. Orders are selected by AIC-minimizing grid search over $p, q \in \{0, 1, 2\}$ and $P, Q \in \{0, 1\}$ (36 combinations). The optimal specification is $\text{SARIMA}(2, 1, 2)(0, 1, 1)_{12}$, with AIC $= 3470.46$ and BIC $= 3502.03$ (Table 6).

Table 6: Top 5 SARIMA Models by AIC (Grid Search)

p	d	q	P	D	Q	s	AIC	BIC
2	1	2	0	1	1	12	**3470.46**	3502.03
2	1	2	1	1	1	12	3472.46	3509.30
2	1	1	0	1	1	12	3474.44	3500.76
2	1	1	1	1	1	12	3476.44	3508.02
1	1	2	0	1	1	12	3477.64	3503.96

Bold = selected model.

3.6.2. SARIMAX

SARIMAX extends SARIMA$(2,1,2)(0,1,1)_{12}$ with the four Granger-selected exogenous variables:

$$\Phi_P(B^s)\,\phi_p(B)\,\Delta^d\Delta_s^D\,y_t = \boldsymbol{\beta}^\top \mathbf{x}_t + \Theta_Q(B^s)\,\theta_q(B)\,\varepsilon_t \tag{3}$$

where $\mathbf{x}_t = (\texttt{winddir}_t, \texttt{humidity}_t, \texttt{sealevelpressure}_t, \texttt{dew}_t)^\top$, standardized using training-set parameters. The fitted model achieves AIC $= 789.87$ and BIC $= 842.49$, substantially lower than SARIMA.

3.6.3. XGBoost

XGBOOST (Chen and Guestrin, 2016) minimizes the regularized objective:

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \quad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda\|\mathbf{w}\|^2 \tag{4}$$

Hyperparameters: max_depth$= 3$, learning_rate$= 0.05$, subsample$= 0.7$, reg_alpha$=$ 0.1, reg_lambda$= 1.5$, min_child_weight$= 5$, with early stopping at 30 rounds (tree_method=hi for efficiency).

3.7. Evaluation Framework

Data are split chronologically into a training set (8 January 2016 – 31 December 2019; $n =$ 1,454; 82%) and a test set (1 January 2020 – 15 November 2020; $n = 320$; 18%) (Figure 7). SARIMA and SARIMAX use rolling one-step-ahead forecasting; at each step t the model is re-fitted on all history $\{y_1, \ldots, y_{t-1}\}$ and produces $\hat{y}_t$ before appending actual y_t. For SARIMAX, 95% prediction intervals are computed at each step using get_forecast().conf_int(alpha=(XGBOOST generates forecasts directly from the feature matrix without re-fitting. Performance is assessed using MAE, RMSE, MAPE, and R^2.

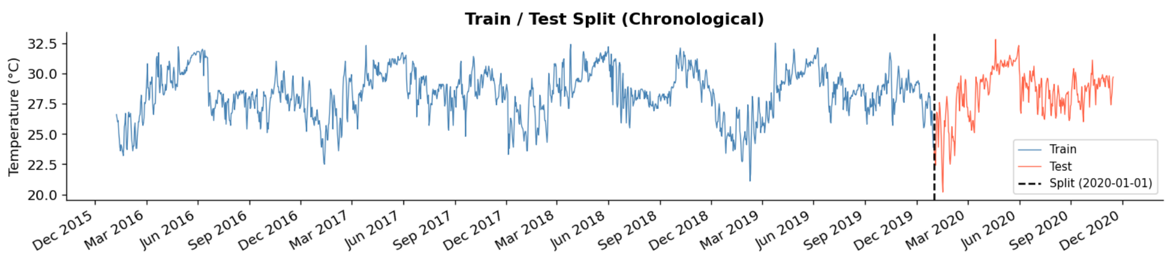


Figure 7: Chronological train/test split at 1 January 2020.

4. Results

4.1. SARIMA Results

The best-fitting SARIMA$(2,1,2)(0,1,1)_{12}$ model is estimated on 1,454 training observations. Residual diagnostics (Figure 8) show approximate white noise beyond lag 3, with minor seasonal autocorrelation at lags 12–13, indicating that the seasonal MA component does not fully capture the annual cycle at $s = 12$. On the test set, SARIMA achieves MAE $= 0.610\,°C$, RMSE $= 0.813\,°C$, MAPE $= 2.22\%$, and $R^2 = 0.836$ (Figure 9).

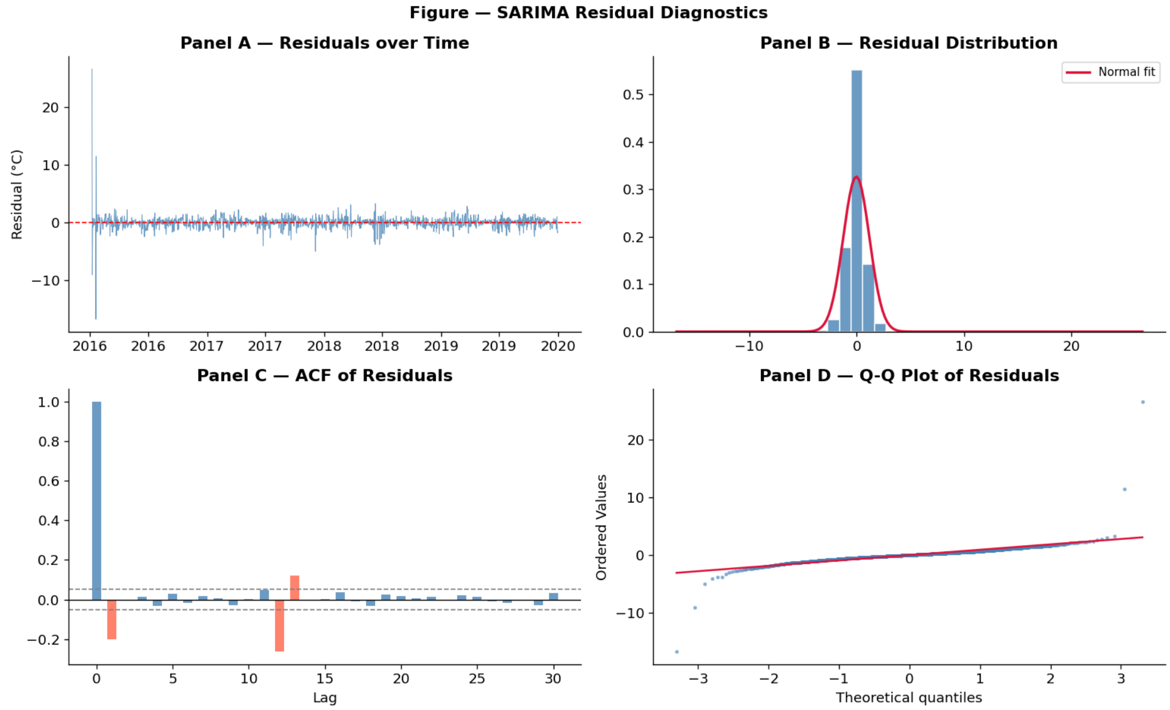


Figure 8: SARIMA residual diagnostics. Panel A: residuals over time. Panel B: distribution vs normal fit. Panel C: ACF of residuals (minor significant spikes at lags 12–13). Panel D: Q–Q plot.

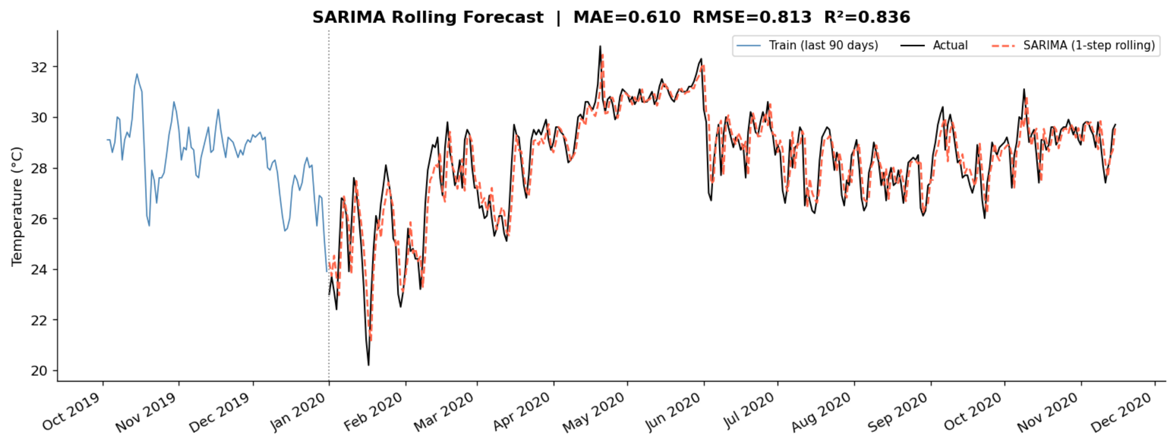


Figure 9: SARIMA rolling forecast vs actual temperature (test set 2020). MAE $= 0.610\,°C$, RMSE $= 0.813\,°C$, $R^2 = 0.836$.

4.2. SARIMAX Results

Incorporating the four Granger-selected exogenous variables substantially improves forecast accuracy. Test MAE falls to $0.231\,°C$ (-62.1%), RMSE to $0.313\,°C$ (-61.5%), MAPE to 0.83% (-62.5%), and R^2 rises to 0.976.

Residual diagnostics (Figure 10) show that SARIMAX residuals are more tightly distributed around zero than SARIMA, with lower spike amplitudes in the early training period. The

ACF of residuals still exhibits significant autocorrelation at lags 12–14, consistent with the $s = 12$ approximation used for both models. The Q–Q plot confirms heavier tails than the normal distribution, driven by initialization spikes in 2016 common to both models.

The 95% prediction interval achieves a coverage rate of **94.7%**, close to the nominal 95% level, confirming that the interval is well-calibrated (Figure 11).

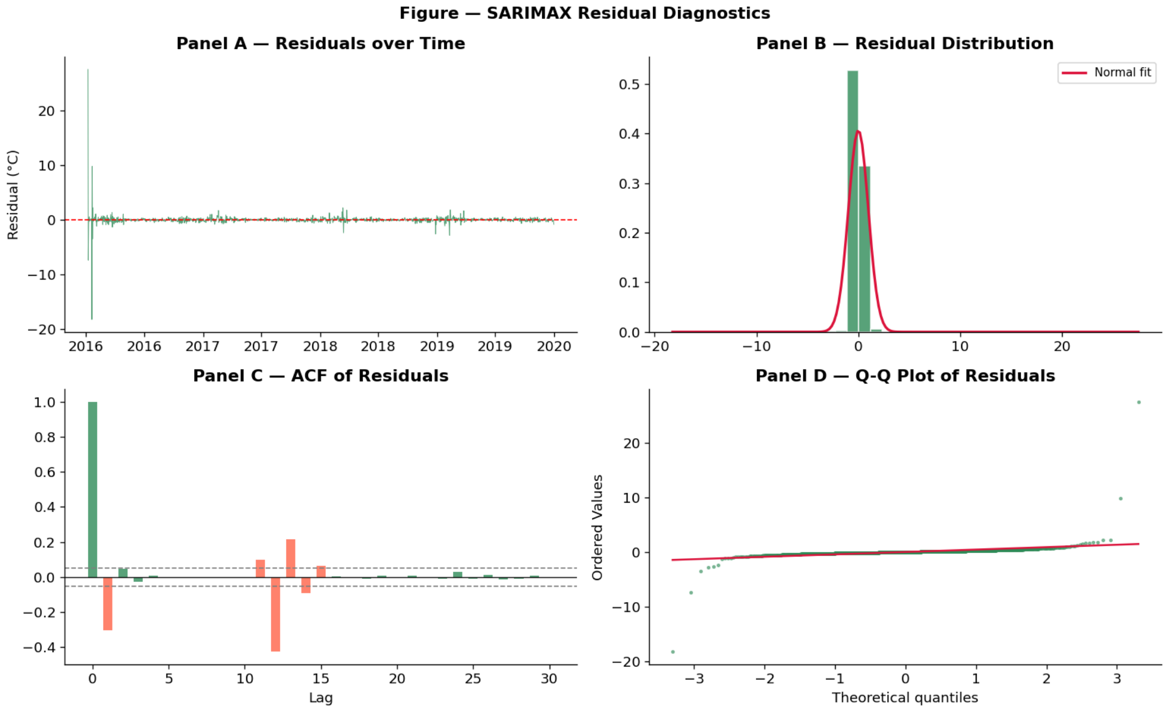


Figure 10: SARIMAX residual diagnostics. Panel A: residuals over time. Panel B: distribution vs normal fit. Panel C: ACF of residuals (significant spikes at lags 12–14 reflect the $s = 12$ seasonal approximation). Panel D: Q–Q plot.

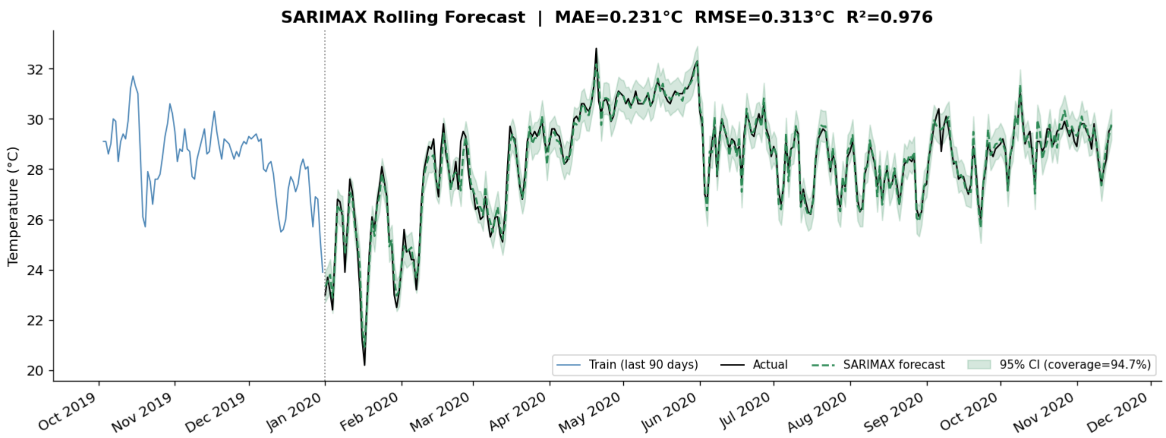


Figure 11: SARIMAX rolling forecast with 95% prediction interval (coverage = 94.7%). MAE = 0.231 °C, RMSE = 0.313 °C, $R^2 = 0.976$.

4.3. XGBoost Results

XGBoost achieves test $R^2 = 0.938$, MAE $= 0.353$ °C, and RMSE $= 0.501$ °C (Figure 12). Feature importance (Figure 13) reveals `temp_lag1` ($\approx 38\%$ gain), `season` ($\approx 25\%$), and `temp_rollmean7` ($\approx 13\%$) as the three most important features.

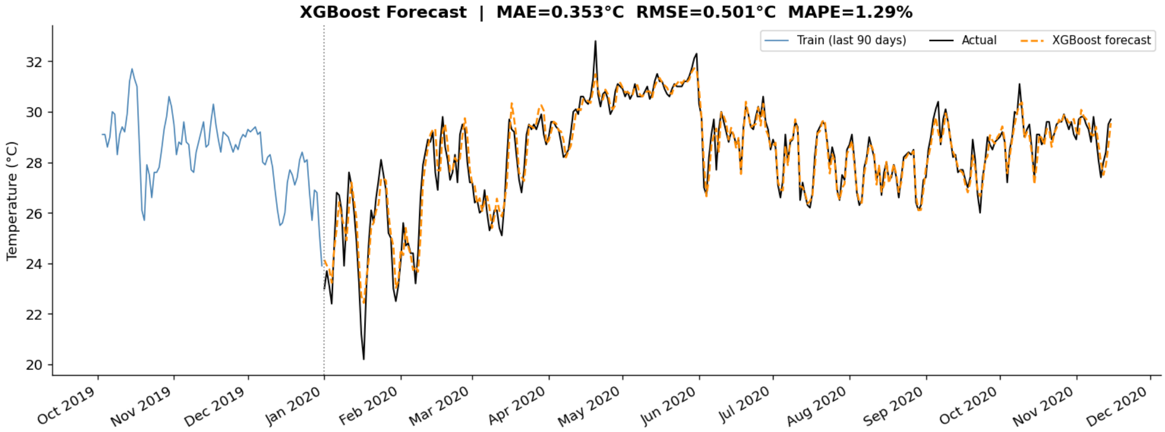


Figure 12: XGBoost forecast vs actual temperature (test set 2020). MAE $= 0.353$ °C, RMSE $= 0.501$ °C, $R^2 = 0.938$.

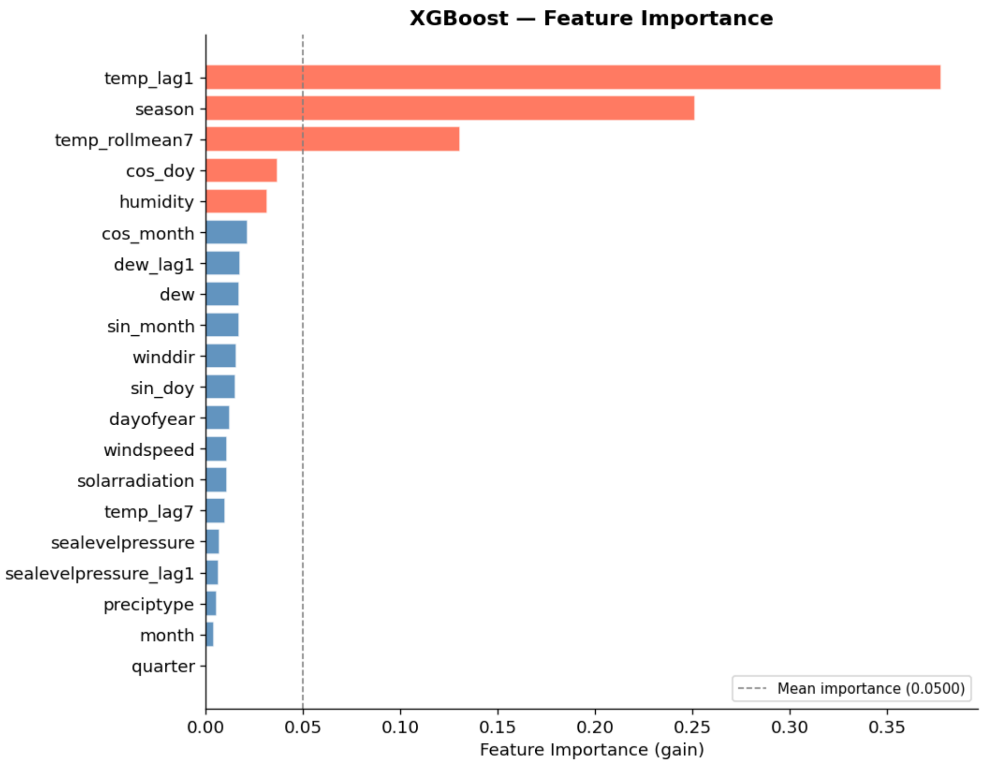


Figure 13: XGBoost feature importance (gain). `temp_lag1`, `season`, and `temp_rollmean7` collectively account for $\approx 76\%$ of total gain.

4.4. Model Comparison

Table 7 summarizes performance across all three models. Figure 14 shows the three forecast series against actual temperature over the test period.

Table 7: Out-of-Sample Forecast Performance (Test Set: Jan–Nov 2020)

Metric	SARIMA	SARIMAX	XGBOOST
Type	Statistical	Statistical	ML
Exogenous vars	None	4 (Granger)	20 features
AIC	3470.46	789.87	N/A
BIC	3502.03	842.49	N/A
Train RMSE (°C)	1.2212	0.9819	0.3533
Test MAE (°C)	0.6100	**0.2310**	0.3532
Test RMSE (°C)	0.8131	**0.3129**	0.5012
Test MAPE (%)	2.2155	**0.8316**	1.2899
Test R^2	0.8362	**0.9758**	0.9378
95% CI coverage	N/A	**94.7%**	N/A
RMSE ratio	0.67	0.32	1.42
Rank	3rd	**1st**	2nd

RMSE ratio = Test RMSE / Train RMSE. Bold = best in column.

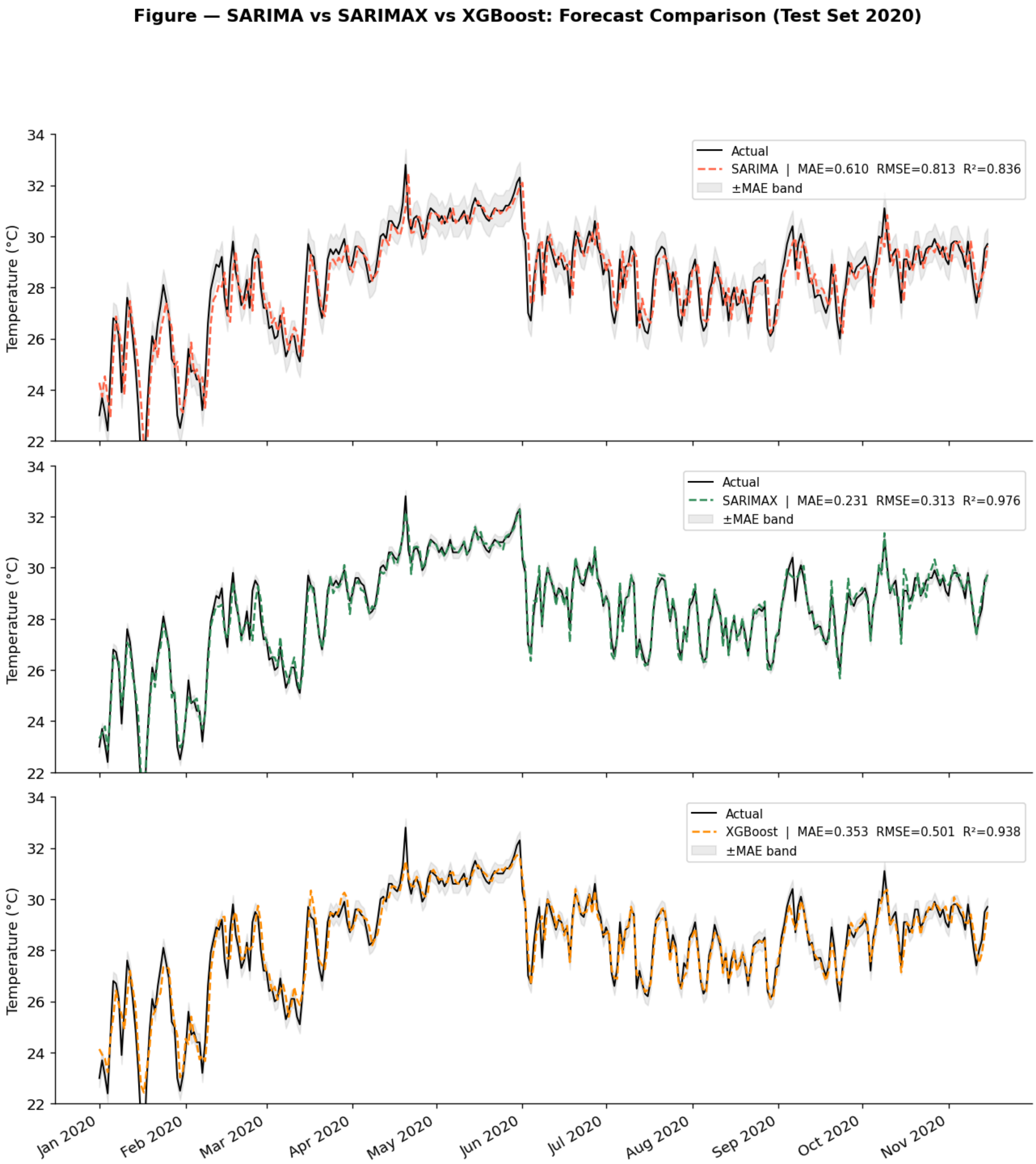


Figure 14: Forecast comparison on the test set (January–November 2020). Each panel shows actual temperature (black) against model forecast (dashed). SARIMAX tracks the observed series most closely.

5. Discussion

5.1. Granger Causality vs Correlation-Based Selection

The most important methodological finding is that Granger causality testing identifies a different and more useful set of predictors than Pearson correlation. Wind direction has only a modest correlation with temperature ($r = 0.23$) but is the most significant Granger-cause ($p < 0.001$, lag 1), suggesting its predictive value operates through a dynamic lagged channel rather than contemporaneous co-movement. Conversely, solar radiation is moderately correlated ($r = 0.22$) but fails the Granger test ($p = 0.58$), implying its association is driven by shared seasonal patterns already captured by the ARIMA structure. An earlier exploratory analysis using only `dew` and `sealevelpressure` — selected by correlation — yielded negligible improvement over SARIMA, confirming that principled variable selection is critical.

5.2. SARIMAX vs XGBoost

SARIMAX outperforms XGBOOST despite using fewer inputs (4 vs 20 features) and no non-linear interaction terms. This suggests that for a target variable with strong autoregressive structure and well-defined seasonal periodicity, a correctly specified statistical model with causally-selected covariates can match or exceed a flexible machine learning approach. XGBOOST's reliance on `temp_lag1` for approximately 38% of its gain further suggests it largely approximates the same autoregressive structure as SARIMA through a different parameterization, without fully exploiting the multivariate information available to SARIMAX.

The well-calibrated 95% prediction interval of SARIMAX (94.7% coverage) provides an additional practical advantage over XGBOOST, which does not natively produce prediction intervals without additional post-processing.

5.3. Limitations

Several limitations should be acknowledged. First, the rolling forecast strategy assumes that exogenous variable values are known at each step; in genuine real-time applications these would need to be forecast independently. Second, SARIMAX optimization produced convergence warnings at a subset of rolling steps; future work should explore alternative optimizers. Third, the $s = 12$ seasonal approximation for daily data is a computational compromise — using $s = 365$ would theoretically be more accurate but is computationally prohibitive for rolling forecast. Fourth, the five-year sample limits the number of full seasonal cycles for estimation. Finally, the analysis does not account for potential structural breaks over longer horizons.

6. Conclusion

This paper compares SARIMA, SARIMAX, and XGBOOST for daily temperature forecasting in Mumbai, India using 1,781 daily meteorological observations from 2016 to 2020. The central contribution is the application of Granger causality testing to select exogenous variables for SARIMAX, identifying wind direction, sea-level pressure, humidity, and dew point as statistically significant predictors.

Incorporating these variables yields a 61.5% reduction in RMSE relative to univariate SARIMA, lifting R^2 from 0.836 to 0.976 and MAPE below 1%. The 95% prediction interval achieves 94.7% empirical coverage, confirming well-calibrated uncertainty quantification. SARIMAX also outperforms XGBOOST ($R^2 = 0.938$), which relies on 20 engineered features, demonstrating that principled variable selection via Granger causality is key to unlocking the full potential of multivariate time series models.

Future research could extend this framework by applying walk-forward cross-validation, testing longer forecast horizons, using $s = 365$ with subsampled rolling forecast, incorporating additional atmospheric predictors, and exploring hybrid SARIMAX–XGBOOST architectures.

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Appendix: Additional Figures

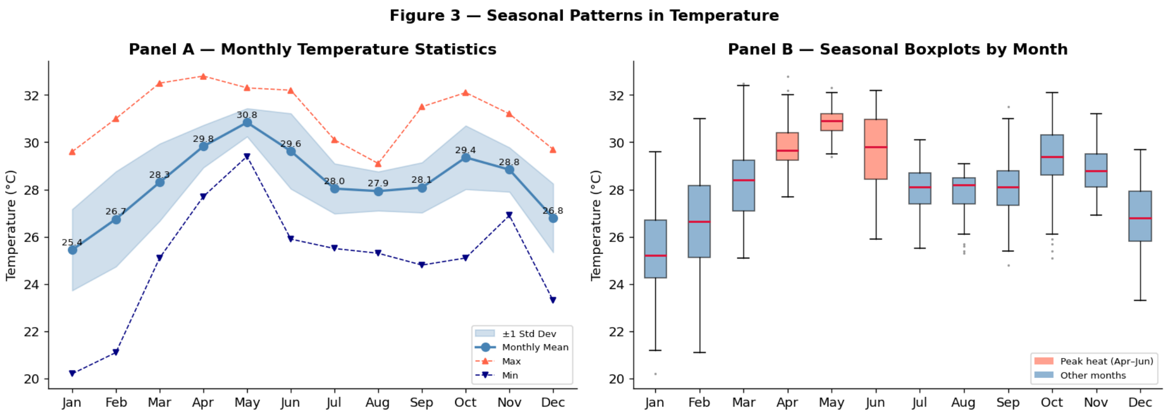


Figure 15: Seasonal patterns. Panel A: monthly mean with ±1 std band. Panel B: boxplots by month; April–June highlighted as peak-heat period.

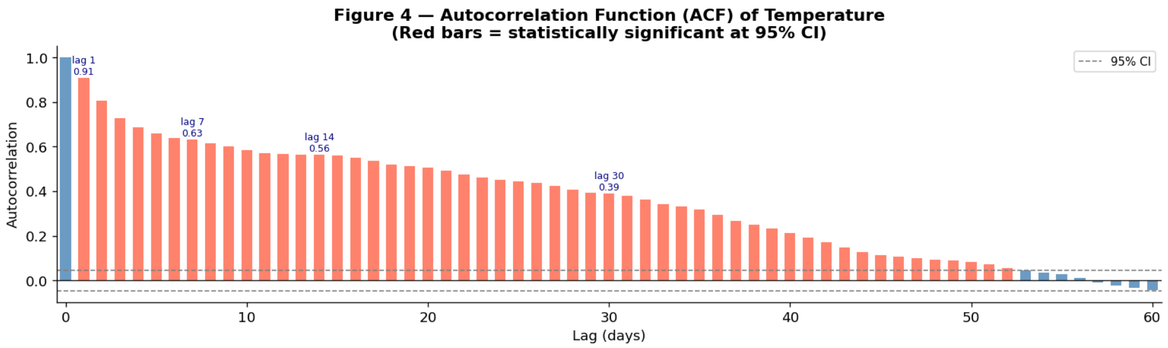
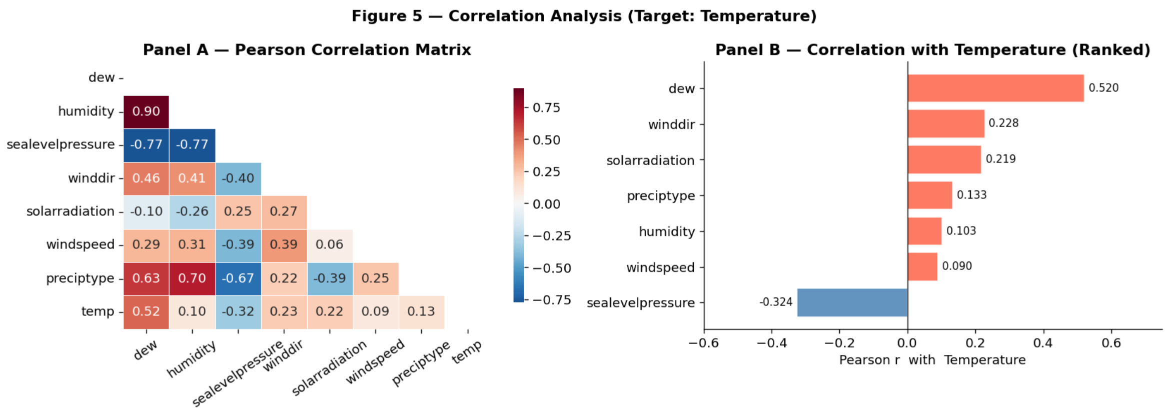


Figure 16: ACF of daily temperature (lags 1–60). ACF remains significant to lag 55, reflecting strong temporal persistence (lag-1 $= 0.91$).



	dew	humidity	sealevelpressure	winddir	solarradiation	windspeed	preciptype	temp
dew								
humidity	0.90							
sealevelpressure	-0.77	-0.77						
winddir	0.46	0.41	-0.40					
solarradiation	-0.10	-0.26	0.25	0.27				
windspeed	0.29	0.31	-0.39	0.39	0.06			
preciptype	0.63	0.70	-0.67	0.22	-0.39	0.25		
temp	0.52	0.10	-0.32	0.23	0.22	0.09	0.13	

Figure 17: Correlation analysis. Panel A: Pearson correlation matrix. Panel B: correlation of each variable with temperature (ranked).

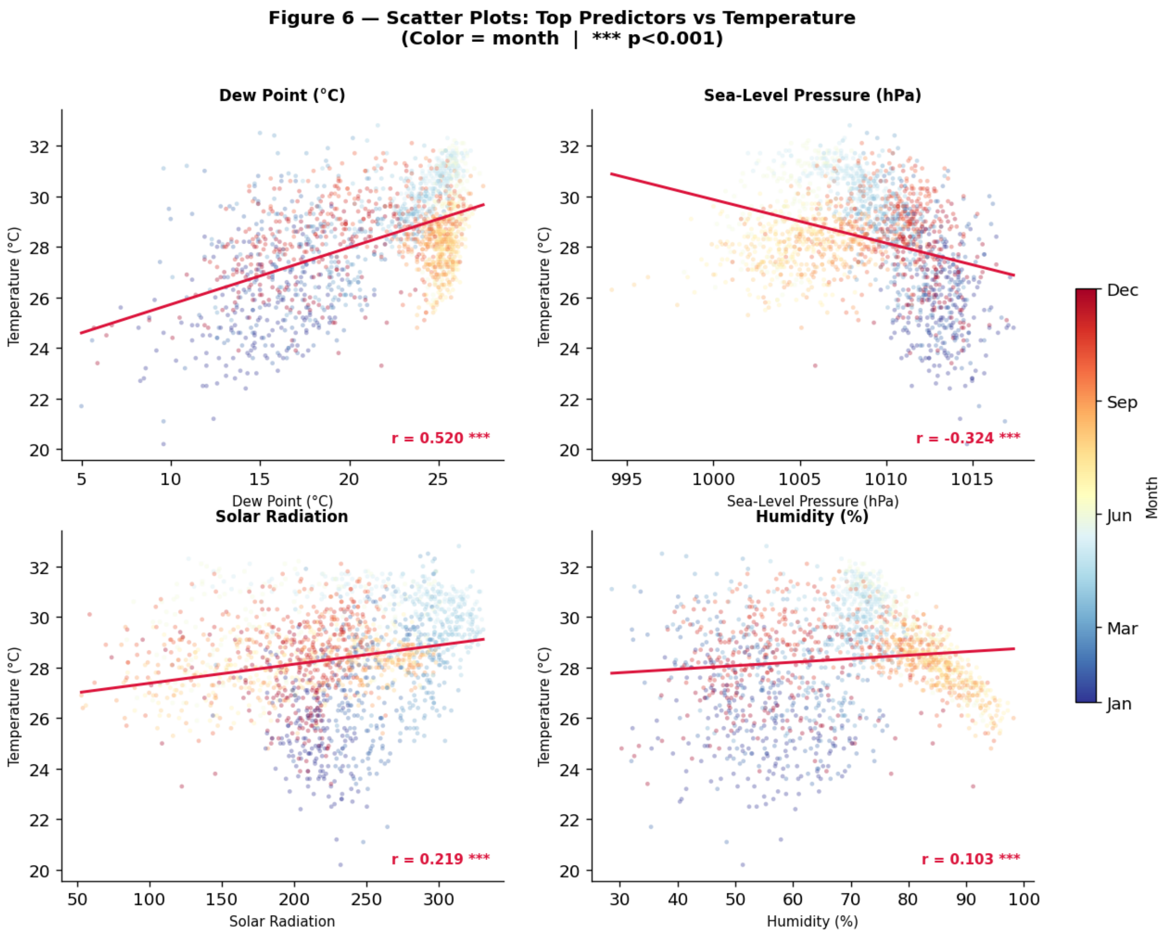


Figure 18: Scatter plots of top predictors vs temperature, colored by month. All correlations significant at $p < 0.001$.

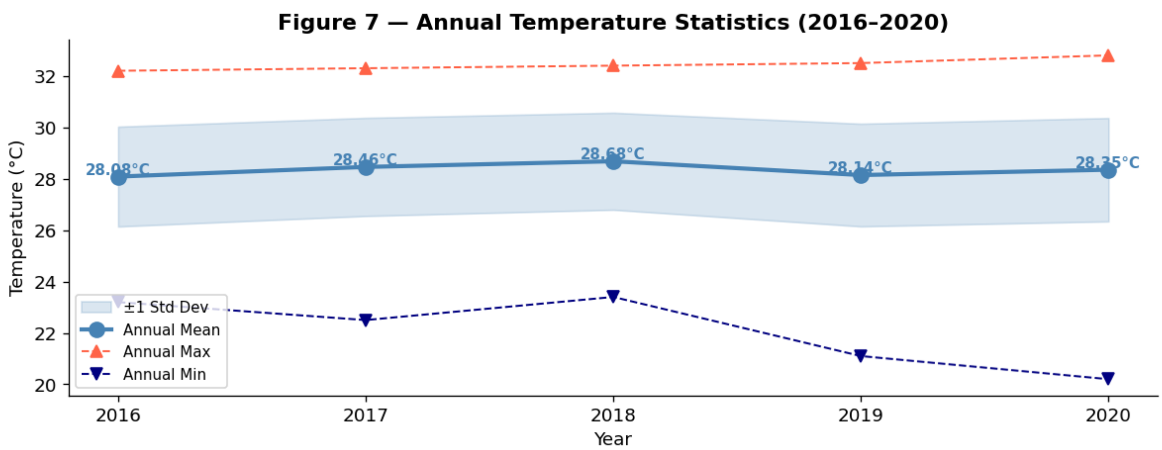


Figure 19: Annual temperature statistics (2016–2020). Annual mean fluctuates narrowly between 28.08 °C and 28.68 °C.